

The Pentagon's Supply Chain: How It's Built, Where It Breaks

A structural map of the U.S. defense industrial base — from five prime contractors down through thousands of sub-tier suppliers to the raw minerals underneath — and what Washington is doing to shore it up.

200,000+

suppliers feed DoD weapons and parts production¹

51 → 5

aerospace/defense prime contractors since the early 1990s¹

~90%

of rare-earth magnet processing runs through China²

\$3.2B

DoD investment in domestic industrial base firms, FY18–24³

01 — STRUCTURE **A pyramid, not a chain**

The defense industrial base is the network of firms, labs, and government facilities that supplies the Department of Defense with everything from fighter jets to routine services⁴. At the top sit five prime contractors — Lockheed Martin, RTX, Northrop Grumman, General Dynamics, and Boeing — who integrate finished weapon systems and collect roughly two-thirds of DoD contract value between them⁵. That concentration is recent: the number of U.S. aerospace and defense primes fell from 51 to 5 since the early 1990s, largely through Pentagon-encouraged consolidation after the Cold War¹.

Below the primes, the base fans out fast. A single missile or aircraft program can run through hundreds of subcontractors — Northrop's work on the MX Peacekeeper missile alone drew on more than 500 subcontractors supplying 19,000 parts⁶. These tier-1, tier-2, and tier-3 firms are usually small, privately held, and thinly capitalized, dependent on a handful of contracts with little ability to pre-finance a production surge⁷. Risk in the base tends to be invisible at the top and concentrated at the bottom: GAO has found DoD has good data on its direct prime contractors but almost none on the lower tiers that actually make specialized parts, and a 2023 effort (DIBMAP) has so far only mapped first- and second-tier suppliers for about 110 weapon programs².

PRIME CONTRACTORS	5 firms · systems integration
TIER 1 SUPPLIERS	major subsystems
TIER 2 / TIER 3 SUPPLIERS	components, castings, electronics
RAW MATERIALS & MINERALS	metals, minerals, chemicals

02 — MATERIALS

The chokepoint is underground

The base of that pyramid runs through China. China controls about 90% of global rare-earth magnet processing and refining, and 98.8% of refined gallium output² — inputs used in precision-guided missiles, fighter jets, submarines, and radar². Pentagon demand for parts built from five critical minerals (antimony, gallium, germanium, tungsten, tellurium) has grown 23.2% a year since 2010; more than 80,000 distinct parts across roughly 1,900 weapon systems now depend on them⁸. As of 2023, about 70% of U.S. rare-earth compound and metal imports still came from China⁹.

Semiconductors carry a related but distinct risk. DoD's own purchasing volume is too small to move the leading-edge chip market, so most military electronics run on mature 28–90nm process nodes rather than cutting-edge silicon¹⁰, verified instead through the Pentagon's Trusted Foundry and Trusted & Assured Microelectronics programs, whose combined funding passed \$1 billion in FY24¹¹.

"America's dependence on China for critical minerals represents a glaring and growing strategic vulnerability."

— Govini, cited in Defense One, April 2025⁸

03 — RESPONSE

Rebuilding, slowly

The Pentagon's main lever is the Defense Production Act, and specifically Title III, its financial-incentive authority — the department's most frequently used DPA tool¹². Since a February 2021 executive order, DoD has obligated over \$893 million in DPA Title III funding across five priority sectors: kinetic capabilities, microelectronics, energy storage, strategic materials, and castings and forgings¹³; combined with other industrial-base programs, total DoD investment reached about \$3.2 billion across 222 awards between FY2018 and FY2024³.

The clearest single bet is on rare earths: the Defense Department became the largest shareholder in MP Materials with a \$400 million equity stake, guaranteeing a 10-year floor price of \$110/kg for the neodymium-praseodymium oxide used in magnets, while helping fund the first U.S. facility to integrate rare-earth mining, separation, and refining¹⁰. On chips, DoD's \$2 billion in CHIPS Act funding built the Microelectronics Commons, a network of eight regional hubs spanning roughly 1,200 universities, companies, and labs across 35 states¹⁴. The Department's 2023 National Defense Industrial Strategy frames the broader goal — resilient supply chains, workforce readiness, flexible acquisition, and economic deterrence — while cataloguing ten "systemic challenges," from sub-tier supplier fragility to funding uncertainty, that none of these programs alone resolve¹⁵.

China's own export controls are forcing the timeline: Beijing has tightened restrictions on rare earths ahead of a planned 2027 U.S. defense ban on Chinese-origin material¹⁶. Analysts are blunt about the gap between ambition and progress — "strategic dependency on China won't be broken by a single investment"¹⁶ — and the same fragility shows up closer to home: tier-2 and tier-3 manufacturers, not the five primes, are now seen as the real constraint on how fast the U.S. can turn defense-budget increases into finished weapons¹⁷.

SOURCES

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